VOICE AGENTIC SMART READER APPLICATION IN SINHALA AND TAMIL FOR VISUALLY IMPAIRED PEOPLE

25-26J-125

ABOUT SOLUTION

Our solution is an AI-powered smart reader designed for visually impaired users who speak Sinhala and Tamil.

It includes a Document Capture and Voice-Guided Navigation component that allows users to capture documents and interact with the content through voice commands. The system then routes the extracted text to language-specific components, including the Sinhala Summarization and Non-Linear Navigation Engine for summarizing and navigating Sinhala documents, and the Tamil Conversational Question Answering and Context-Aware Recommendation component for answering questions and providing relevant content suggestions.

The final output is delivered through an Emotional Text-to-Speech component, which generates expressive speech for a more natural listening experience. Additionally, to address the lack of emotional speech resources for Sinhala, we created a Sinhala emotional TTS dataset and released it publicly on the Hugging Face Hub to support future research in low-resource language speech technologies.

PROBLEM STATEMENT & STAKEHOLDERS

Context:

- Access to printed and digital documents for visually impaired users within Sinhala and Tamil language environments, particularly in education and everyday information access.

Stakeholders:

- Visually impaired Sinhala and Tamil speakers [Primary Users], Teachers, caregivers, and assistive technology providers [Secondary Users], and Educational institutions, accessibility organizations, and government bodies [Decision Makers].

Observed Issue:

- Existing OCR systems produce high recognition errors for Sinhala and Tamil texts due to script complexity and capture conditions, while current Text-to-Speech systems lack natural voice output and contextual awareness. Additionally, most tools do not provide real-time document capture guidance or intelligent document navigation.

Why it Matters:

- These limitations prevent visually impaired users from independently accessing educational and informational content, increasing reading time, errors, and overall barriers to learning and daily information access.

RESEARCH GAP & OBJECTIVES

1.

- Existing assistive reading technologies focus on separate tasks like OCR, screen reading, or TTS, lacking a fully integrated intelligent reading system for visually impaired users.
- Current systems provide limited interaction features such as voice-based navigation, conversational question answering, and personalized recommendations.
- Sinhala and Tamil support remains limited, especially for emotional speech synthesis, structured document understanding, and intelligent navigation.
- Most screen readers rely on linear reading with neutral speech, reducing user engagement and usability.

2.

- To develop a voice-agentic smart reader that enables visually impaired users to capture, understand, navigate, and interact with documents in Sinhala and Tamil.
- To integrate document capture, conversational QA, personalized recommendations, voice navigation, and emotional TTS into a unified accessibility platform.

SOLUTION OVERVIEW

Our system is an AI-powered Smart Reader for visually impaired Sinhala and Tamil users, enabling voice-based interaction with documents through intelligent navigation, conversational AI, and expressive speech synthesis.

1.Real-Time Document Scanning & OCR

Guides users to align documents and extracts structured text using OCR while preserving the document layout.

2.Intelligent Navigation & Summarization

Text is cleaned and segmented into sections. A machine learning–based intent classification model detects voice commands (jump, search, read, summarize) to enable non-linear document navigation.

3.Conversational Interaction & Smart Recommendation

A RAG-based QA system allows users to ask questions during reading, while a reinforcement learning mechanism recommends relevant content based on user interests.

4.Emotion-Aware Text-to-Speech

Emotion classification models detect the emotional tone of the text and guide the TTS system to generate expressive speech output.

SYSTEM ARCHITECTURE

Intelligent Document Capture System
IT22326454

- Provides real-time voice guidance for document alignment
- Uses semantic segmentation for document detection
- Performs OCR using Google Vision API

Emotional Text-to-Speech System
IT22336590

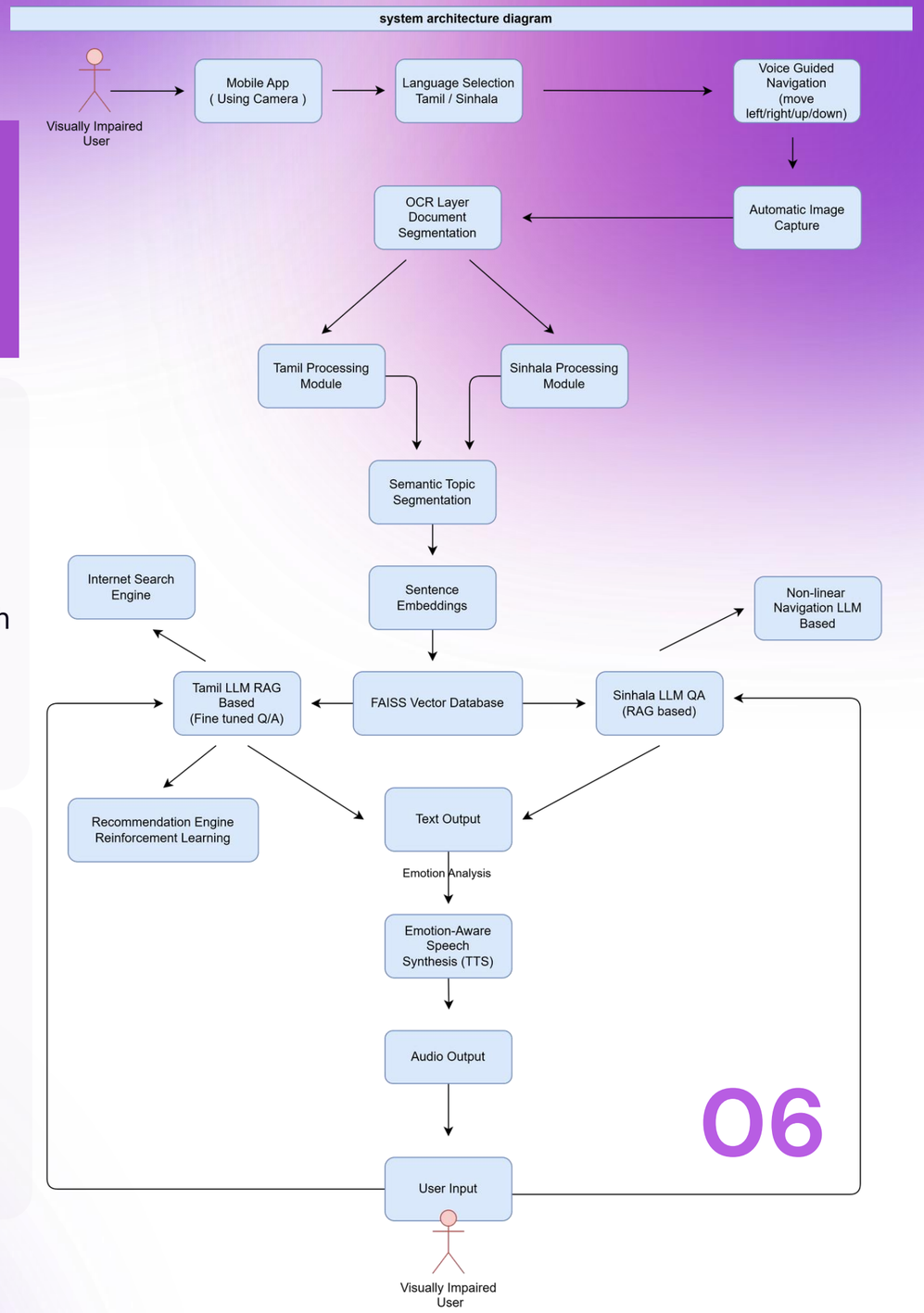
- Performs Sinhala and Tamil emotion classification
- Generates emotionally expressive speech using MiniTTS

Conversational QA & Recommendation System
IT22348234

- Handles user questions using RAG pipeline
- Retrieves relevant article content using FAISS vector search + Tamil LLM
- Provides personalized news recommendations using reinforcement learning

Intelligent Navigation & Summarization
IT22316486

- Enables voice-based navigation of documents
- Uses FAISS semantic retrieval for content access + Sinhala LLM
- Generates summaries and supports non-linear document exploration



COMMERCIALIZATION

The proposed AI-powered Smart Reader can be commercialized as an assistive technology platform that improves accessibility for visually impaired and low-vision users, particularly for low-resource languages such as Sinhala and Tamil.

Target Users

- Visually impaired and low-vision individuals
- Students accessing academic documents
- News readers and digital content consumers
- Government accessibility programs and public libraries

Product Opportunities

- Mobile Accessibility Application – Voice-controlled smart reader for documents and news content
- Educational Platform Integration – Accessibility support for universities, schools, and digital libraries
- Media and News Platforms – Personalized audio news reading and interactive content exploration
- Developer APIs – Emotion-aware TTS, document navigation, and multilingual NLP services

Revenue Model

- The system follows a mobile application subscription model.
- The Normal (Free) version provides basic OCR-based document reading and linear audio playback.
- The Premium (Paid) version offers advanced AI features such as non-linear navigation, AI-based summarization, conversational question answering, and personalized content recommendations.

INTEGRATION

Integration Points

- Google Cloud Vision API – Document OCR and text extraction
- RunPod Hosted Tamil LLM API – Handles Tamil question answering and RAG responses
- HuggingFace Sentence Transformers – Generates semantic embeddings
- FAISS Vector Database – Stores document embeddings for semantic retrieval
- Speech-to-Text Engine – Converts user voice queries into text
- Emotional TTS – Converts generated text responses into emotional expressive speech

Data Formats:

- Image frames (camera input)
- Extracted text (UTF-8 strings)
- Vector embeddings (numerical arrays)
- JSON API responses between modules

Deployment / Runtime Setup

- Mobile Android Application – Handles camera capture, user interaction, and voice guidance
- RunPod Cloud GPU – Hosts the fine-tuned Tamil LLM model for question answering
- Railway Backend Server – Manages API routing, RAG pipeline, and system integration
- Render Cloud Services with huggingface hub – Hosts supporting backend services
- Local Development via Ngrok – Used by team members to expose local modules for integration testing
- FAISS Vector Database – Runs locally within the backend for fast semantic retrieval

Overall Architecture: Hybrid deployment (Mobile + Cloud APIs + Local services)

DEMO FLOW

Language Selection - **Home Screen**

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Real-Time Document Scanning & OCR- **M C Aluwihare**

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Emotion-Aware Speech Reading - **B J A Siriwardana**

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Intelligent Navigation & Summarization - **A D Hewagama (sinhala)**

Conversational Interaction & Smart Recommendation - **M Sharumathan(Tamil)**

KEY FEATURES AND METRICS

1) Document Capture and Voice-Guided Navigation System

- Captures document images, extracts text using OCR, and enables visually impaired users to navigate the content through voice commands.

2)Emotional Text-to-Speech (TTS) System

- Converts generated text into emotionally expressive speech using language-specific emotion classification and TTS synthesis model. Additionally, a custom Sinhala emotional TTS dataset.

3)Sinhala Summarization and Non-Linear Navigation Engine

- Generates concise Sinhala summaries and allows users to access document sections through voice-driven non-linear navigation.

4)Tamil Conversational Question Answering and Context-Aware Recommendation System

- Uses a RAG-based LLM pipeline to answer user questions and recommend relevant content based on learned user preferences.

THANK YOU

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